

Parth N. Kheni

pkheni@bu.edu | 908-335-5448 | github.com/parthnkheni | linkedin.com/in/parthnkheni

EDUCATION

Boston University

Boston, MA

B.S. Computer Engineering, Concentration in Machine Learning

Expected Dec 2026

- Related Coursework: Machine Learning, Probability Statistics & Data Science, Computational Linear Algebra, Applied Algorithms, Cloud Computing, Software Engineering, Computer Architecture, Differential Equations

Rutgers University

New Brunswick, NJ

B.S. Computer Engineering | GPA: 3.97 | Dean's List

Sept 2023 – May 2024

SKILLS

Programming: Python, SQL, MATLAB, C++ (primary), Java, C#, Objective-C, OOP

Libraries/Frameworks: Numpy, Matplotlib, Pandas, scikit-learn, TensorFlow, PyTorch

Cloud/Tools: Azure AD, GitHub, Microsoft 365, Arduino, Android Studio, API, Power BI, WorkManager

Machine Learning: Predictive Modeling, Statistical Modeling, Time-Series Analysis, Anomaly Detection

EXPERIENCE

Beth Israel Deaconess Medical Center (BIDMC), Machine Learning Intern

Sept 2025 – Present

- Lead a multi-institutional collaboration between BIDMC and Johns Hopkins to develop and integrate a learning-based autonomous wound closure algorithm within the da Vinci Research Kit (dVRK) platform for future clinical translation, coordinating requirements and experiment milestones across teams.
- Build repeatable wound-closure experiments by standardizing task setup, ROS data capture, labeling guidelines, and evaluation metrics; maintain consistent dataset formats across runs to enable clean comparisons and faster iteration on autonomy behavior.
- Maintain a reproducible NVIDIA Isaac Sim + ORBIT-Surgical pipeline on BU's HPC Cluster using Singularity containers, and document protocols, results, and meeting action items to keep development aligned.

Boston University Center for Space Physics, Machine Learning Researcher

Apr 2025 – Present

- Build end-to-end Python pipelines for POES and OMNI time-series data, generating model-ready datasets with 7M+ 2-second satellite samples per month by ingesting NetCDF files, interpolating gaps, enforcing L-shell and MLT quality cuts, and joining with 4-hour solar wind history windows.
- Train and evaluate PyTorch neural networks and Ridge regressors to forecast energetic electron precipitation (E4_0/E4_90), improving RMSE and correlation over prior baselines using standardized features and time-ordered train/val/test splits.
- Improve model generalization across storm conditions and radiation belt locations by evaluating temporal resolution trade-offs, quantile-based flux aggregations, and local-time features, with focus on rare high-flux event performance.

PUBLICATIONS & PRESENTATIONS

- Capannolo, L., Pettit, J., Elliott, S., Morales, J. P. B., Bibeau, A., **Kheni, P. N.** *Feb 2026*
“Monitoring radiation belt electrons at LEO: preliminary results of MAPLE (MAPs of LEO Electrons) model.”
COSPAR 2026 – Abstract (submitted)

PROJECTS

LLM-Assisted Incident Summarization & Clustering (Microsoft-affiliated), Applied ML Engineer *Jan 2026 – Present*

- Build an end-to-end incident intelligence pipeline: ingest large-scale system/application logs, generate structured incident summaries with LLMs, embed summaries, and cluster incidents to identify duplicates and failure patterns.
- Implement a backend service and lightweight UI/API to explore incidents and clusters, making practical accuracy vs cost/latency tradeoffs to support scalable operation on noisy real-world logs.

Whack-a-Mole Reaction Game on FPGA (Nexys4 DDR, Verilog), FPGA Design Engineer

Sept 2025 – Dec 2025

- Built a real-time Whack-a-Mole game on a Nexys4 DDR FPGA with 30-second multi-difficulty rounds, 5 LED “moles”, and per-mole switch/hammer input, using Verilog for debounced I/O, random selection, and top-level design.
- Improved reliability and integration, delivering a glitch-free demo by designing input and game-control modules, coordinating interfaces across a 4-person team, and iterating with Vivado simulation, timing analysis, and on-board testing.

Personal Finance App, Android Developer

Jan 2025 – May 2025

- Built an Android personal finance app in Android Studio (Java) using Retrofit/Gson + Finnhub.io for real-time quotes (RecyclerView, MPAndroidChart); added SharedPreferences portfolio persistence, MainActivity → BrowseStocksActivity navigation, and WorkManager + Notifications for price alerts.

AWARDS & HONORS

NSF ACCESS / Jetstream2 Allocation (CIS260066) – \$31,444 awarded; 146,000 GPU SUs.

Jan 2026 – Present

GEM Workshop – Invited Speaker (Portland, ME) – selected to present machine learning research.

July 2026

ACTIVITIES

BU Baja Society of Automotive Engineers, Head Engineer of Computer Systems

Sept 2024 – Present

BU Jalwa Dance Troupe, Secretary

Sept 2024 – Present